Aditya (Adit) Venkataraman

Ph.D. Student in Geography | University of Utah
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A. EDUCATION

University of Utah, Salt Lake City, UT *Expected August 2028*

Ph.D. in Geography

Advisors: Dr. Simon Brewer and Dr. Summer Rupper. Research focus: building machine learning emulators for global glacier change.

Northwestern University, Evanston, IL *December 2017*
M.S. in Analytics

Carnegie Mellon University, Pittsburgh, PA *May 2016*
B.S. in Statistics and Machine Learning

B. RESEARCH, TEACHING, AND PROFESSIONAL EXPERIENCE

Research and Academic Experience

University of Utah, Salt Lake City, UT *August 2023-Present*
Graduate Research Assistant

- Building machine learning emulators for global glacier change.

Teaching Experience

University of Utah, Salt Lake City, UT *Spring 2026*

- Lab Section Instructor, GEOG 3050: Problem Solving in Physical Geography
- Fieldwork Teaching Assistant for Dr. McKenzie Skiles, Intro to Geospatial Field Methods

Carnegie Mellon University, Pittsburgh, PA *2015-2016*

- Teaching Assistant, 36-225: Introduction to Probability Theory

Industry Data Science Experience

NielsenIQ, Chicago, IL *January 2018-August 2023*

Senior Data Scientist, North America Data Science Operations

- Automated and productionized a regression model that tracks ethnic proportions of sales for 50,000 brands.
- Developed a Python sampling algorithm to select from 9,000 stores for imputing store characteristics.
- Analyzed and provided timely insight on data differences for a critical $10 billion platform-transition project.
- Enhanced market-profile maps for more than 820 critical retailer trade areas for internal stakeholders.
- Enhanced and automated a program that updates consumer-platform retailer trade-area market profiles quarterly.
- Co-led development of client inquiry tools supporting raw-data, product-level, and market-level inquiries.
- Led technical-server transitions for program migrations, server administration, and user setup.
- Provided methodology enhancements, ad hoc analysis, and subject-matter expertise on perishable random-weight items.
- Developed programming interview questions and trained approximately 300 Emerging Leader/Technologist Program associates.

HERE Technologies, Chicago, IL *September 2017-December 2017*

Data Science Consultant

- Analyzed temporal and geospatial data to increase Automatic Closure Detection coverage by 68% for German roads.
- Developed a Python program to identify highway-like clusters from arterial roads using hierarchical clustering and dynamic time warping.

NASA Jet Propulsion Laboratory, Pasadena, CA *June 2017-August 2017*

Data Science Intern

- Developed and implemented a heuristic-optimization algorithm in Python that reduced Amazon Web Services spending by more than 63%.
- Documented results in a public repository and presented results to senior leadership; recommendations were used to reduce cloud-storage costs across the organization.

Shopify, Ottawa, Canada *October 2016-June 2017*

Data Science Industry Practicum

- Analyzed a 40 GB dataset to determine important success predictors for customer conversion.

- Developed and presented a Flask data-visualization application to provide insights on customer conversion.

C. PUBLICATIONS AND PRESENTATIONS

Manuscripts in Preparation

Venkataraman, A., Brewer, S., and Rupper, S. "*Analyzing a Small-Scale Foundation Model as an Emulator for Global Glacier Mass Balance*." Manuscript in preparation for Annals of Glaciology; planned submission November 2026.

Venkataraman, A., Brewer, S. C., McOmber, H., and Codding, B. F. "*Optimizing Invasive Species Control: Removing Russian Olive in the Bear River Basin to Maximize Streamflow to the Great Salt Lake*." Manuscript in preparation for Journal of Environmental Management; planned submission November 2026.

Conference Posters

Venkataraman, A., Brewer, S., and Rupper, S. "*Quantifying Global Glacier Mass Change Using Surrogate-AI Methods*." Poster presented at the AGU Fall Meeting, 2025.

Venkataraman, A., Brewer, S., and Rupper, S. "*Quantifying Global Glacier Mass Change Using Surrogate-AI Methods*." Poster presented at the American Association of Geographers Annual Meeting, 2025.

Venkataraman, A., Brewer, S., and Rupper, S. "*Analyzing a Small-Scale Foundation Model as an Emulator for Global Glacier Mass Balance*." Forthcoming poster presentation at the AGU Fall Meeting, 2026.

Invited Talks and Workshops

Venkataraman, A. "*Understanding the Role of Artificial Intelligence in Geographical Problems*." Invited talk, University of Utah, April 2024.

Venkataraman, A. "*Surrogate Modelling Workshop*." Invited workshop leader, Spatial Utah Data Science, University of Utah, October 2026.

D. PROFESSIONAL DEVELOPMENT AND MEMBERSHIPS

GlaMacLeS Glaciology and Machine Learning Summer School *July 4-13, 2026*
Taft-Nicholson Center, Centennial Valley, Montana

- Attended an intensive summer school on machine learning and artificial intelligence for glaciology, including deep-learning emulation of ice physics and student project development.

American Geophysical Union
Member

E. TECHNICAL SKILLS

Python: Pandas, TensorFlow, Matplotlib, scikit-learn, SciPy, NumPy, PySpark, Dask, Plotly, Seaborn, GeoPandas, and Shapely
R: ggplot2, dplyr, and Shiny
Databases and GIS: Oracle, IBM Netezza, PostgreSQL, PostGIS, and ArcGIS
Machine Learning: supervised learning, unsupervised learning, natural language processing, and deep learning
Other: Apache Spark, Amazon Web Services, Bash scripting, technical writing, Git, Tableau, and Slurm